

Project Testing

FitBuddy AI — Project Deliverable

Create a clean testing document covering functional verification, API verification, automated tests, and performance-testing preparation without inventing measured results.

Deliverables in this single phase document

- Functional Test Cases
- API Test Coverage
- Performance Testing Plan

Evidence status — Test cases and commands are documented; no performance metrics or actual pytest results were supplied.

6.1 Functional Test Cases

- Home page loads and presents the profile form.
- Profile fields accept the intended values and the form can be reused.
- Generate My Plan produces a 7-day plan and nutrition/recovery tip.
- Result page allows practical feedback submission.
- Updated-plan view reflects the submitted feedback.
- Invalid input is rejected through the documented validation layer.

6.2 API Test Coverage

The documented JSON endpoints are the health check, plan generation, feedback submission, and user retrieval endpoints. The source also identifies pytest + httpx for automated application and HTTP testing.

Endpoint	Test purpose
GET /api/health	Verify service/status response
POST /api/generate	Verify plan generation from submitted profile data
POST /api/feedback	Verify feedback submission and revised-plan behavior
GET /api/users	Verify user retrieval with appropriate protection

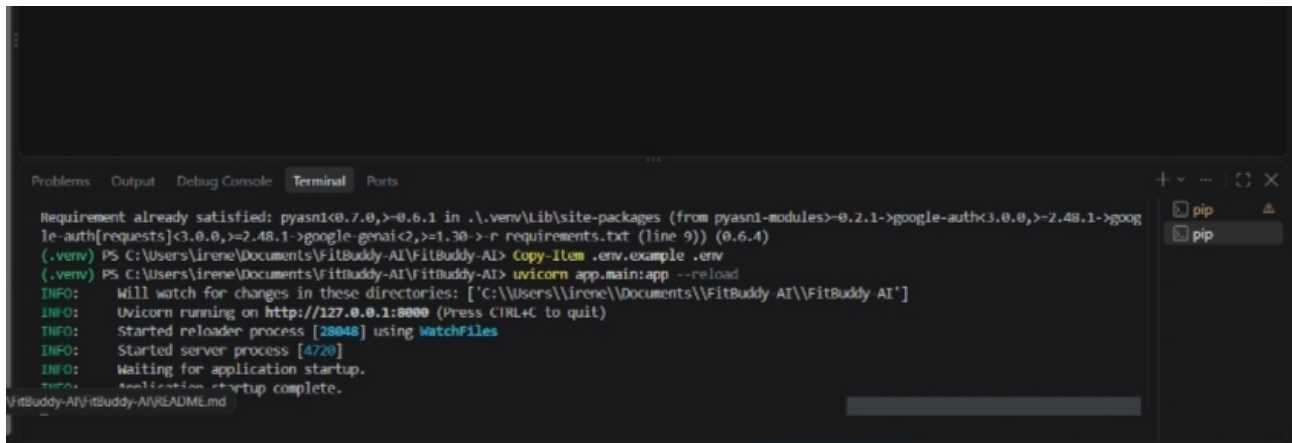
6.3 Automated Testing

The README instructs the project team to run **pytest -q** from the project root with the virtual environment active. The supplied documentation does not include actual terminal output proving a particular test result, so no pass count is asserted here.

6.4 Performance Testing Plan

- Measure response time for /api/health, /api/generate, and /api/feedback under representative load.
- Record throughput, error rate, and resource usage during a controlled test.
- Separate demo-mode measurements from Gemini-enabled measurements because external model calls can affect latency.
- Record actual values only after running the performance test.

Project Evidence / Screenshots



```
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in .\venv\Lib\site-packages (from pyasn1-modules>=0.2.1->google-auth<3.0.0,>=2.48.1->google-auth[requests]<3.0.0,>=2.48.1->google-generativeai<2,>=1.30->r requirements.txt (line 9)) (0.6.4)
(.venv) PS C:\Users\Irene\Documents\FitBuddy-AI\FitBuddy-AI> copy-item .env.example .env
(.venv) PS C:\Users\Irene\Documents\FitBuddy-AI\FitBuddy-AI> uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\Irene\\Documents\\FitBuddy AI\\FitBuddy AI']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [28048] using WatchFiles
INFO: Started server process [4720]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Screenshot evidence: local Uvicorn server running. Use actual pytest/performance output as final evidence.

Source note: This phase document was converted from the supplied FitBuddy AI project documentation, README, configuration/dependency information, and supplied project screenshots. Where the source did not provide a measured result, named team member, or actual date, the document does not present one as a completed fact.